

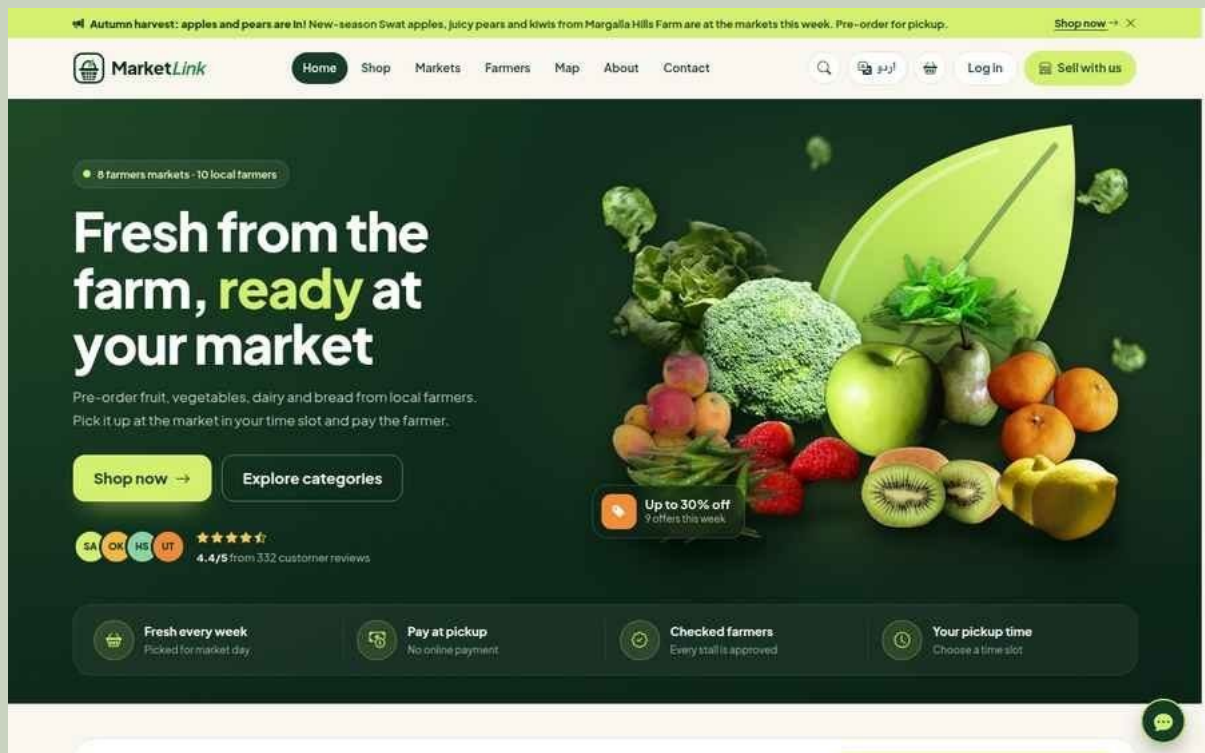
MarketLink

TECHWIZZ 7 | 2026
End-to-End Web Solutions

PROJECT REPORT

Bringing local farmers markets online with pre-orders, live maps and an AI assistant.

Cash on pickup · English & Urdu · Theme: eGreen Basket



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ISSUED

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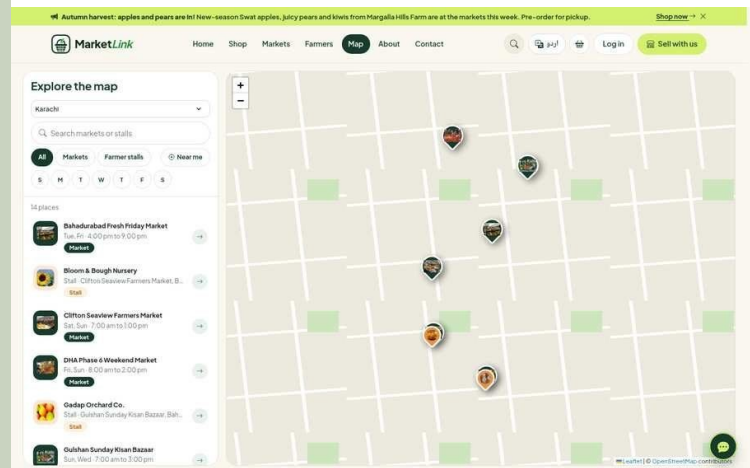
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SECTION

01

Introduction

Weekend farmers markets in Pakistani cities still run entirely in person. MarketLink puts them online without changing how people pay: order ahead, collect at the stall, pay the farmer in cash.



1.1 Problem definition

WHAT GOES WRONG EVERY WEEKEND

Farmers markets in urban Pakistani cities, such as the weekend stalls in Clifton, DHA or F-9 Park, have no online presence. Every week the same problems repeat for shoppers, farmers and organisers:

01

No visibility before the market

Customers travel without knowing which farmers are coming, what they bring, the prices, or what is already sold out.

02

Produce goes to waste

Farmers guess how much to harvest. Too much spoils unsold; too little means lost sales.

03

Early-morning sell-outs

Popular produce such as Sindhri mangoes is gone early, and families who arrive later go home empty-handed.

04

No market management

There is no central place for opening hours, stall maps, verified reviews or complaints.

1.2 The MarketLink solution

THEME: EGREEN BASKET

MarketLink connects local growers with city shoppers. Shoppers find markets on an interactive map, check stock by weight, reserve produce with a pre-order for a pickup time slot, and pay the farmer in cash at pickup. There are no delivery fees and no online payment step.

- **Farmers** publish their weekly stock, prices, offers and pickup windows.
- **Customers** reserve real stock with a pre-order and collect it at a time slot they choose.
- **Administrators** approve farmers, manage markets and categories, moderate content and read the platform reports.

“ *Live stock that updates as orders come in, protection against clashing pickup slots, a full English and Urdu interface, and an assistant that answers in Roman Urdu.* ”

KEY DELIVERABLE

1.3 Team & mentor

WHO BUILT WHAT

MEMBER	ROLE	OWNED
Muhammad Saad	Team lead & Backend	Node.js REST API, security, JWT sign-in, e-mail services
Ghulam Mustafa	Frontend & UI/UX	React app, SCSS design system, animations
Fahad Musa	Database & QA lead	MongoDB schema, indexes, seed data, testing
Muhammad Hamza	Maps & AI	Leaflet maps, Roman Urdu chatbot, documentation
Sir Sufyan	Project mentor & advisor	Technical guidance and architecture review

1.4 Project objectives

FIVE THINGS MARKETLINK HAD TO DO

1. Let customers browse markets, farmers and products by location, day, category and price, with a map and directions to the pickup point.
2. Let customers pre-order against the farmer's real stock for a pickup slot, then track, change or cancel the order before the farmer's cut-off time.
3. Give farmers a simple dashboard for products and weekly stock, offers, pickup windows, pre-orders, sales insights and replies to reviews.
4. Give the administrator control over users, markets, categories, content, announcements and reports.
5. Work well on phones, in English and Urdu, with secure, role-based access.

1.5 Scope

IN, OUT, AND HOW SUCCESS WAS MEASURED

IN SCOPE	OUT OF SCOPE (SRS 1.5)
Customer, farmer and admin areas with role-based access	Online payment (payment is made at pickup)
Markets, farmers, products, categories and cities	Delivery or courier logistics (pickup only)
Pre-orders with pickup slots and cut-off times	Checking farmer identity, licences or organic certificates
Reviews, ratings, favourites and restock reminders	Native mobile apps (the website is responsive instead)
Maps (OpenStreetMap), directions, notifications and e-mail	
AI assistant in English, Urdu and Roman Urdu	

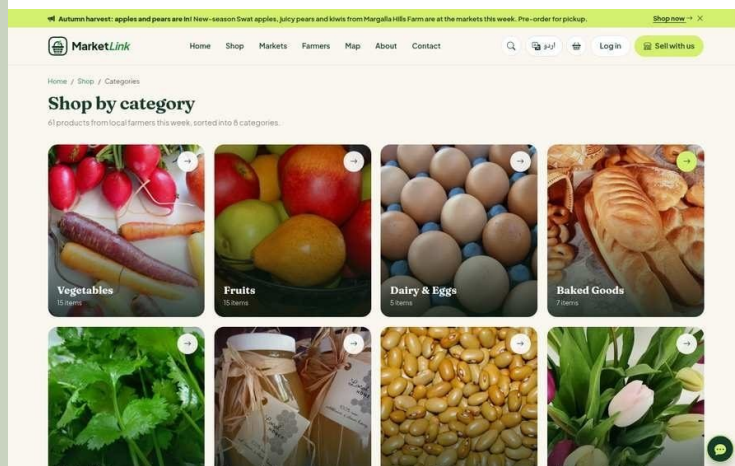
SUCCESS MEASURE	RESULT AT HAND-IN
Every requirement of SRS 1.6 has a page or screen	Yes: listed one by one, with verification, in Appendix B
Business rules enforced by the server, not only the screens	Stock, cut-off, slot capacity, pickup clash and role checks are covered by automated tests (Section 6.2)
Works on phones and computers in English and Urdu	Checked at 18 screen widths (320 to 1920 px) in both languages
Ready to run with demo data	One folder with the built app, packages, sample data and documents

SECTION

02

Project overview

One platform, three kinds of user. Each role gets its own portal, and every page works on phones, tablets and computers in English and Urdu.



2.1 Who uses MarketLink

FOUR ROLES

ROLE	WHAT THEY DO ON MARKETLINK
Guest	Browses markets, farmers and products, uses the map and the assistant, and can check out without signing up first: an account is created and the password is e-mailed.
Customer	Browses markets on the map, shops by weight (250 g, 500 g, 1 kg ...), pre-orders for a pickup slot, pays cash at the stall, writes reviews, keeps favourites and family sharing, and reports "I did not receive it" problems.
Farmer	Runs a stall page, manages Products & Pricing, accepts or declines orders, marks orders ready, replies to reviews, and gives a reason when they cannot attend a market.
Administrator	Approves farmers, manages markets and categories, decides pickup problems, tracks market attendance, and runs 17 platform reports with CSV export and print.

2.2 Key features

TEN THINGS WORTH SHOWING THE JUDGES

M Live market map Leaflet and OpenStreetMap show every market with directions and opening days.	P Pre-order & pickup slots Customers choose a pickup window; a customer cannot book pickups at two markets at the same time.
G Buy by weight Farmers sell kg or litre products in smaller steps with a minimum and a maximum per order.	C Price comparison The product page shows the same produce at other stalls first, e.g. "Rs 30 less per kg".
U English & Urdu Full right-to-left Urdu pages and a "Write in Urdu" helper next to every Urdu box.	A Roman Urdu assistant A chatbot answers shopper questions in English, Urdu and Roman Urdu from live data.

R

Reports & analytics

All 17 platform reports on one page, with CSV download, print and saved copies.

!

Pickup problem handling

Customer, farmer and admin each have a flow to resolve an order that was not received.

D

Market-day check-in

Farmers check in or say why they cannot come; customers see a live "at the market today" badge.

S

Weekly stock template

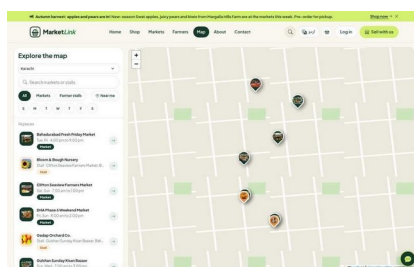
A recurring template refills the stall each market day, and low-stock alerts warn the farmer.

2.3 Screens at a glance

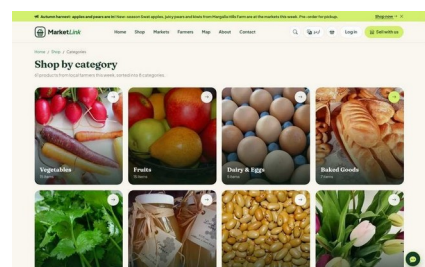
THE FIRST PAGES A SHOPPER SEES, THEN THE BUYING JOURNEY



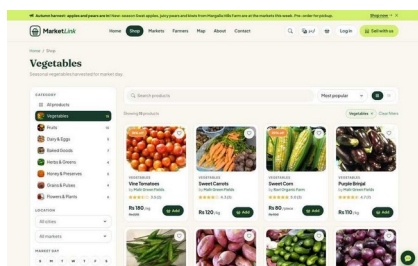
A Landing page



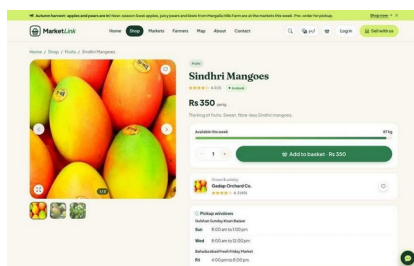
B Market map



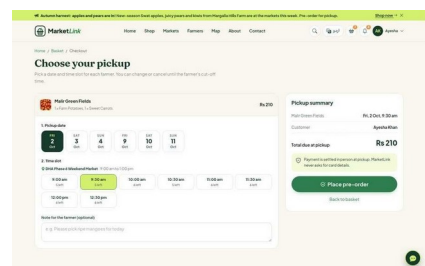
C Categories



D Shop & filters



E Product & weight



F Checkout & slot

2.4 What was delivered

THE HAND-IN PACKAGE

77
PAGES

148
API ENDPOINTS

20
COLLECTIONS

1,122
AUTOMATED CHECKS

DELIVERABLE	DETAILS
The MarketLink website	77 pages in three areas (public and customer, farmer, admin), in English and Urdu
REST API	148 endpoints under /api with input checks, role checks and rate limits
Database	MongoDB with 20 collections, a schema script with validation rules and indexes, and sample data
Demo data	The seed script creates 21 users, 8 markets, 62 products, 435 pre-orders, 341 reviews and more (Appendix E)

DELIVERABLE	DETAILS
Documentation	This report, the test plan and test report, the ReadMe, and design notes in the docs folder
Automated tests	410 API checks, 712 browser checks and page sweeps at 18 screen widths in both languages
Ready-to-run package	One folder with the built app, the server packages, screenshots of every page and the reports

2.5 Roles & permissions

CHECKED ON THE SERVER FOR EVERY REQUEST

Without a login the API answers 401; with the wrong role it answers 403.

CAPABILITY	GUEST	CUSTOMER	FARMER	ADMIN
Browse markets, farmers, products and the map	✓	✓	✓	✓
Use the AI assistant	✓	✓	✓	✓
Check out (guests get an account by e-mail)	✓	✓	-	-
Track, change and cancel own orders before the cut-off	-	✓	-	-
Favourites, family sharing, reviews, restock reminders	-	✓	-	-
Manage products, weekly stock, offers and pickup windows	-	-	✓	-
Accept, decline, mark ready and complete pre-orders	-	-	✓	-
Reply to reviews; check in on the market day	-	-	✓	-
Approve or suspend farmers; activate or deactivate customers	-	-	-	✓
Manage markets, cities and categories	-	-	-	✓
Moderation queue, announcements, offer banner, FAQs	-	-	-	✓
Place an order for a customer; decide pickup problems	-	-	-	✓
17 reports, farmer rankings, CSV export and print	-	-	-	✓

2.6 Project activities

SEVEN STAGES, BUILT IN SHORT ROUNDS

STAGE	WHAT WE DID
1. Study and planning	Read the SRS, listed every requirement, chose the MERN stack, drew the architecture, data flow and flowcharts, designed the database
2. Foundations	Set up the server and the React app, accounts and login for every role, the database models and the demo data
3. Customer features	Markets and map, farmers, products with filters and search, basket, checkout with pickup slots, orders, favourites, reviews
4. Farmer features	Registration wizard and approval, products and weekly stock, inventory, offers, pickup windows, market day, pre-orders, sales insights
5. Admin features	Dashboard, user management, markets and cities, categories, moderation, announcements, banner, messages, reports and rankings
6. Extras	Urdu with right-to-left layout, the AI assistant, restock reminders, receipt confirmation, notifications, e-mail, SEO
7. Testing and polish	API and browser tests, responsive sweeps, alignment checks, defect fixes, screenshots, documentation

SECTION

03

Design specifications

MarketLink is a MERN application (MongoDB, Express, React, Node.js) built in short rounds: each round shipped working features, was tested, and fed the next one.

3.1 Approach

HOW THE WORK WAS RUN

- **Requirement checklist.** Every line of SRS section 1.6 became a row in a checklist (Appendix B) with the page or screen that meets it, checked again after every round.
- **Short rounds.** Plan a small set of features, build them, test them, review them, then move on, so problems were found early.
- **Test first for the important rules.** Stock reservation, cut-off times, pickup slots and role checks are enforced on the server and covered by automated tests.
- **Real data from day one.** The demo data covers every situation: a farmer waiting for approval, a suspended farmer, sold-out products, orders in every status.
- **AI as an assistant.** Claude Code helped with coding, tests and the first draft of the documents; every decision and check stayed with the team (Appendix D).

3.2 Technology stack

WHAT WE USED, AND WHY

LAYER	TECHNOLOGY	WHY WE CHOSE IT
Frontend	React 19, Vite, React Router 7, Bootstrap 5 with a custom SCSS theme, Bootstrap Icons, Recharts, DataTables, Leaflet, intl-tel-input	Fast pages, responsive design system, right-to-left Urdu support, charts and tables for the dashboards
Backend API	Node.js 20+, Express 5, Mongoose, JWT, bcrypt, Helmet, Nodemailer, an hourly scheduler	REST controllers, role checks on every route, input cleaning, e-mail and background jobs
Database	MongoDB (Community Server or Atlas), Mongoose	20 collections with validation rules and map (geo) indexes
Maps & AI	Leaflet, OpenStreetMap, OSRM routes, Google Maps links; rule-based assistant, optional Claude for text	Market maps and directions without a paid key; Roman Urdu chatbot
Caching	Redis (optional)	Used when configured; in-memory cache otherwise
Testing	Playwright (Chromium), Node.js API scripts, ESLint	Full journeys, responsive sweeps and API checks after every round

3.3 System architecture

THREE TIERS AND FOUR OUTSIDE SERVICES

The browser runs the React app, the Express server exposes a REST API and serves the built app, and MongoDB stores the data. Map tiles, routes and e-mail come from outside services.

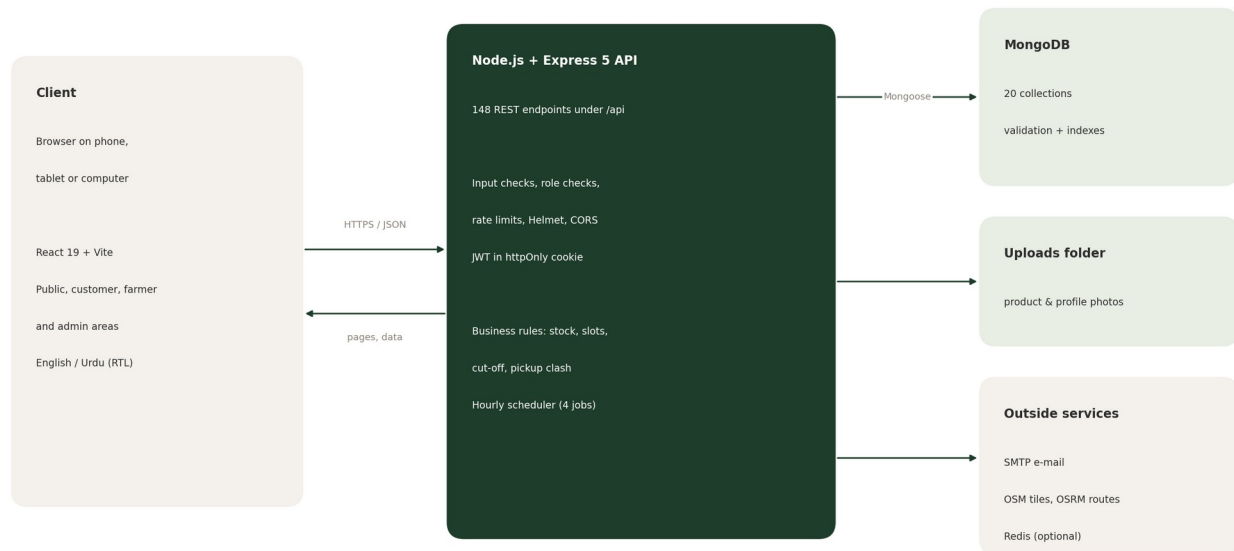


Figure 1 High-level system architecture

LAYER	WHAT IT DOES
Presentation	The React app: public pages, the customer area, the farmer portal and the admin dashboard, in English and Urdu
Application	The Express API: input checks on every request, role checks on every protected route, rate limits on login, chat and AI features
Security	Passwords hashed with bcrypt, the login token in an httpOnly cookie, Helmet security headers, CORS, file-type checks on uploads
Data	MongoDB through Mongoose, with validation rules and indexes on every collection
Services	E-mail (SMTP), the hourly scheduler (weekly stock template, low-stock alerts, "market open today" notices, offers that end on their last day), maps and routes

3.4 Business rules enforced on the server

THE RULES THAT MAKE PRE-ORDERING TRUSTWORTHY

RULE	HOW IT WORKS
Stock reservation	Each item is reserved with one atomic database update that fails when the stock is not enough; the order is created only after the reservation. If one farmer's part of a checkout fails, every earlier reservation is released and the earlier orders are removed.
Pickup slot validation	The date must match one of the farmer's pickup windows, must not be a closed date, the slot must be free (capacity per slot) and the time must be before the farmer's cut-off.
No pickup clash	Pickups on the same day at different markets must start at least 60 minutes apart. Pickups at the same market may share a time. The check covers the checkout and the customer's other open orders.
Checkout limits	At most 10 farmers in one checkout; note field limited to 500 characters; one order per farmer is created.
Order numbers	Human-friendly and unique, in the form ML-YYMMDD-XXXX.

RULE	HOW IT WORKS
Modify an order	Only while the order is placed or accepted and before the cut-off. Old stock is released and the new amount reserved; if that fails the original reservation is restored.
Cancel an order	Stock returns to the farmer, the movement is logged, and both the farmer and the customer are notified.
Receipt confirmation	After completion the customer answers "Did you receive your order?" (30 days). "No" opens a pickup problem and tells the farmer and admins; repeated reports reopen it.
Guest checkout	Name, e-mail, phone, address and the Terms are required. If the e-mail already has an account the customer is asked to log in instead.
Restock reminders	One request per product and e-mail address; refused for products that are in stock.

3.5 REST API overview

148 ENDPOINTS UNDER /API, GROUPED BY WHO MAY CALL THEM

ROUTE GROUP	PURPOSE	ACCESS
/auth	Register, login, logout, forgot and reset password, quick account for guest checkout, profile	Public / logged in
/markets, /farmers, /products, /categories, /cities, /map, /search	Catalogue, stalls, map data and live search	Public
/orders, /customer	Pre-orders, receipts, reorder, family orders, dashboard, favourites, household	Customer
/reviews, /notifications, /flags	Reviews, bell menu and notifications, reports of inappropriate content	Logged in
/farmer	Products, weekly stock, offers, inventory, pickup windows, market day, orders, insights	Approved farmer
/admin	Farmers, customers, markets, cities, categories, moderation, banner, reports, rankings, pickup problems	Admin
/assistant, /ai	Chat assistant and text helpers ("Write with AI", Urdu writer)	Rate limited
/newsletter, /contact, /faqs, /banners, /announcements, /credits	Newsletter, contact form, FAQs, offer banner, seasonal notices, photo credits	Public

3.6 Security & data protection

EIGHT MEASURES

MEASURE	HOW IT IS DONE
Passwords	Hashed with bcrypt; reset links are one-time and valid for 30 minutes
Sessions	JSON Web Token stored in an httpOnly cookie, so page scripts cannot read it
Role checks	Middleware checks the role on every protected route; farmers are also locked until the admin approves them
Rate limits	Failed logins: 20 per 15 minutes; sign-up and contact forms: 40 per 15 minutes; chat and AI helpers: 30 per minute
Headers and CORS	Helmet security headers and a CORS policy; compressed responses
Uploads	File-type and size checks on every image upload; files are stored in the uploads folder
Input and data rules	Input cleaning on every request, protection against NoSQL injection, MongoDB validation rules and unique indexes
Consent	The Terms & Conditions checkbox is required on both sign-up forms and the accepted version is stored

3.7 Notifications, e-mail & background jobs

WHO HEARS ABOUT WHAT

Every important event creates an in-app notification (the bell menu) and, where it matters, a branded HTML e-mail sent with Nodemailer. Without SMTP settings the e-mails are printed in the server window, so the demo works with no setup.

EVENT	WHO IS TOLD	HOW
Pre-order placed (with pickup details and directions link)	Customer and farmer	In-app + e-mail
Accepted, ready, declined, completed	Customer	In-app + e-mail
Order changed or cancelled by the customer	Farmer (and customer for cancel)	In-app
Low stock or sold out	Farmer	In-app + e-mail
Product back in stock ("Remind me")	Customers and guests who asked	In-app + e-mail
Farmer cannot come to the market	Customers with a pickup that day	In-app + e-mail (once)
Not received / not received again	Farmer and admins	In-app (+ e-mail when repeated)
Markets open today; new farmer approved	Customers of that city or saved markets	In-app

An hourly scheduler runs four jobs: it refills the weekly stock template, ends offers after their last day and puts the usual price back, sends the "market open today" notice an hour before the first market opens, and raises the low-stock alerts.

3.8 Urdu interface & AI assistant

LANGUAGE AS A FEATURE

Urdu. The whole site can be switched to Urdu with a right-to-left layout, the Noto Nastaliq Urdu font, Urdu product names, FAQs, terms and server messages. A script scans every text in the app and reports missing translations (0 missing at hand-in). Farmers and the admin get a "Write in Urdu" helper next to every Urdu box.

Assistant. A rule-based assistant answers from live data in English, Urdu and Roman Urdu, remembers follow-up questions, and keeps a saved chat history for signed-in users. Text helpers such as "Write with AI" use Claude when an API key is set and a built-in writer otherwise.

QUESTION	LANGUAGE	ANSWER
which farmer is highly rated	English	The top-rated farmer with review count, and a top-5 list
tamatar kahan milega	Roman Urdu	Tomatoes with price, farmer and pickup days
sab se sasti sabzi	Roman Urdu	The five cheapest vegetables in stock
any offers this week?	English	Products on offer with the percent off
markets open now	English	Markets open now, or the ones opening later today
is Malir Green Fields at the market today?	English	Whether the farmer is there now, later, has left or is away
password bhool gaya	Roman Urdu	How to reset the password, with a link
سب سے اچھی ریٹنگ والا کسان کون ہے؟	Urdu	The top-rated farmer, answered in Urdu

3.9 User interface design

LOOK, FEEL AND ACCESS

- **Look and feel.** Dark green on a cream background, white cards and real photos of produce, markets and farmers. Fraunces for headings, Plus Jakarta Sans for text and Noto Nastaliq Urdu for Urdu.

- **Home page.** A banner with the headline, "Shop now" and "Explore categories", then sections that alternate between dark green and cream: categories, the offer banner, top picks, a video tour, top-rated farmers, markets and map, the farm-to-table story, customer reviews, why choose us and the FAQ.
- **Phones.** A bottom navigation bar on every page, dialogs that open as bottom sheets, filters that slide in from the side, tables that turn into cards.
- **Accessibility.** Keyboard access to every control, visible focus, alt text on every image, reduced motion respected, colour contrast checked.

SECTION

04

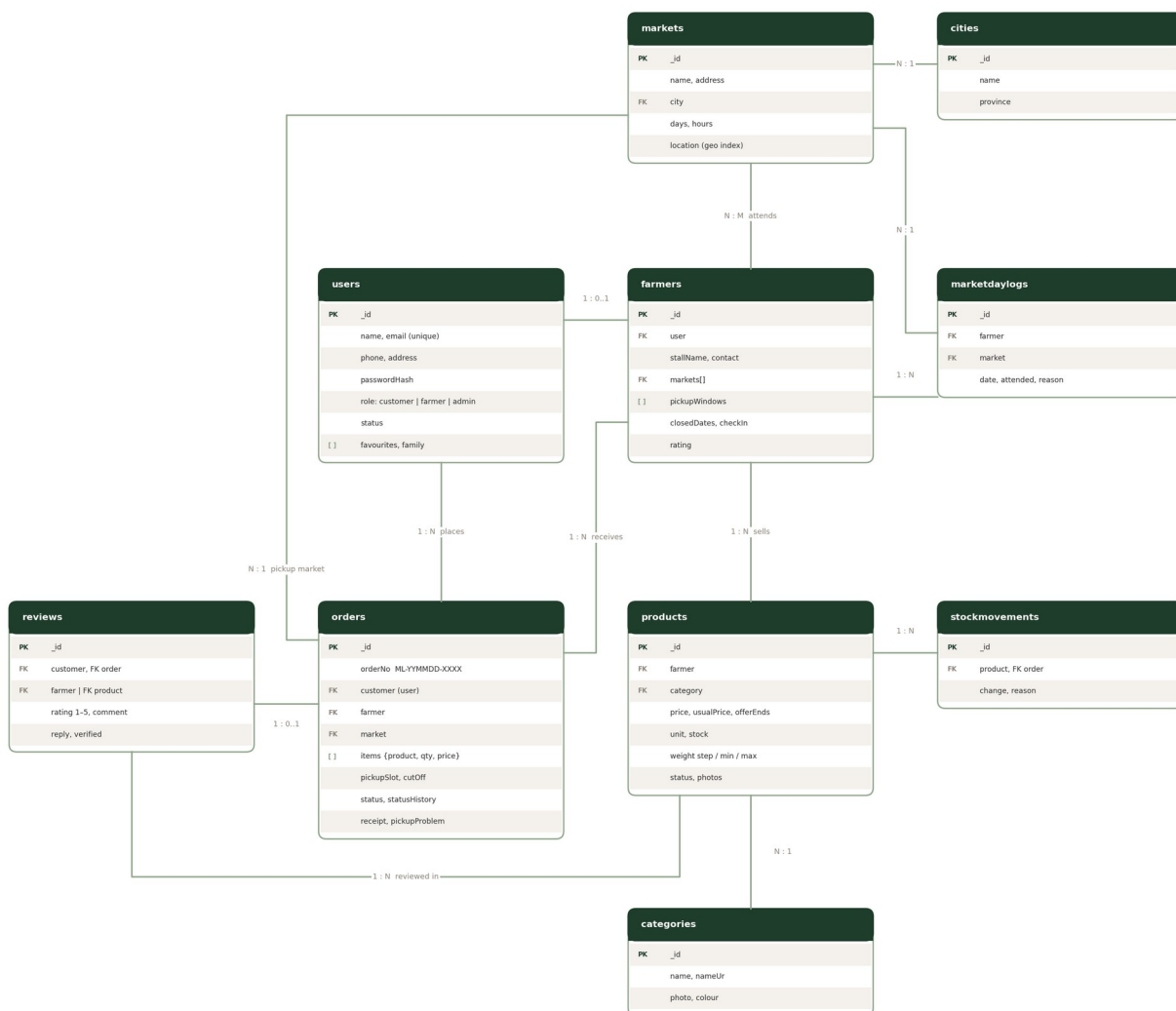
Database design

MongoDB stores the data in 20 collections. Relations are stored as references, except the order items and the pickup windows, which live inside their order or farmer document.

4.1 Entity relationship diagram

THE TEN CORE COLLECTIONS

Each card is a collection. PK marks the primary key, FK a reference to another collection, and [] an array stored inside the document. The labels on the lines give the cardinality: one user places many orders, one order has at most one review.



PK primary key FK reference to another collection [] embedded array lines read "left/top entity : right/bottom entity".

The other ten collections (notifications, announcements, sitebanners, contactmessages, contentflags, faqs, reports, subscribers, restockrequests, assistantchats) are listed in Table 4.1.

Figure 2 Entity relationship diagram of the core collections

ALL 20 COLLECTIONS

COLLECTION	WHAT IT STORES
users	Every login: customers, farmers and administrators (role, status, favourites, family)
farmers	The stall profile of a farmer: markets, days, pickup windows, closed dates, today's check-in, rating
markets	Farmers markets with address, days, hours and map position
cities	The list of cities for markets and farmers
categories	Product categories with an Urdu name, photo and colour
products	A farmer's product: price, usual price and last day of an offer, unit, stock, how it is bought (by weight, smallest and largest amount), status, photos, SEO fields
orders	Pre-orders with items, pickup slot, status history, receipt answer and pickup problem
marketdaylogs	One line per farmer and market day: came, or could not come and why
reviews	Ratings and comments for farmers and products, with the farmer's reply
stockmovements	The history of every stock change (orders, restocks, template, corrections)
notifications	In-app messages for each user
restockrequests	"Remind me when it is back" requests (unique per product and e-mail)
announcements	Platform notices, optionally for a season
sitebanners	The home page offer banner the admin edits
contactmessages	Contact form messages with their topic
contentflags	Reports of inappropriate content for the moderation queue
faqs	Questions and answers in English and Urdu
reports	Saved platform reports
subscribers	Newsletter subscribers
assistantchats	Chat history and memory of signed-in users

4.2 Database scripts

MONGODB INSTEAD OF .SQL

WHY THERE IS NO .SQL FILE

MongoDB does not use SQL. The project ships a mongosh schema script that creates the collections with validation rules and indexes, the demo data as JSON files for MongoDB Compass or mongoimport, and a seed command that fills a fresh database with the same data. The ReadMe says the same.

SECTION

05

Flowcharts & data flow

The diagrams show the system in plain steps: who sends what, where it is stored, and what happens to a pre-order from the basket to the review.

5.1 Data flow diagram, level 0

THE CONTEXT: WHO TALKS TO MARKETLINK

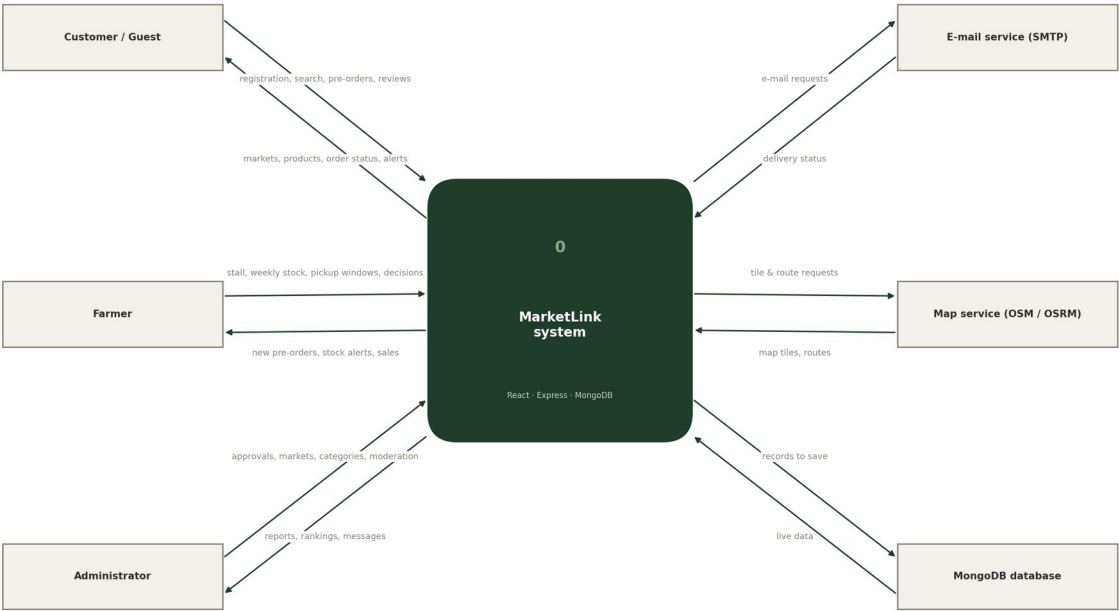


Figure 3 Data flow diagram, level 0 (context)

WHO	SENDS TO MARKETLINK	RECEIVES FROM MARKETLINK
Guest visitor	Searches, browsing, a guest pre-order	Markets, farmers, products and the map
Customer	Registration, pre-orders, reviews, receipt answers	Order status, notifications, recommendations
Farmer	Stall details, weekly stock, pickup windows, order decisions	New pre-orders, alerts, sales reports
Administrator	Approvals, markets, categories, moderation, the offer banner	Platform reports, rankings, messages
E-mail, map, database	Requests for e-mails, map tiles and routes; data to save	Delivered e-mails, tiles and routes, live data

5.2 Data flow diagram, level 1

EIGHT PROCESSES AND SIX DATA STORES

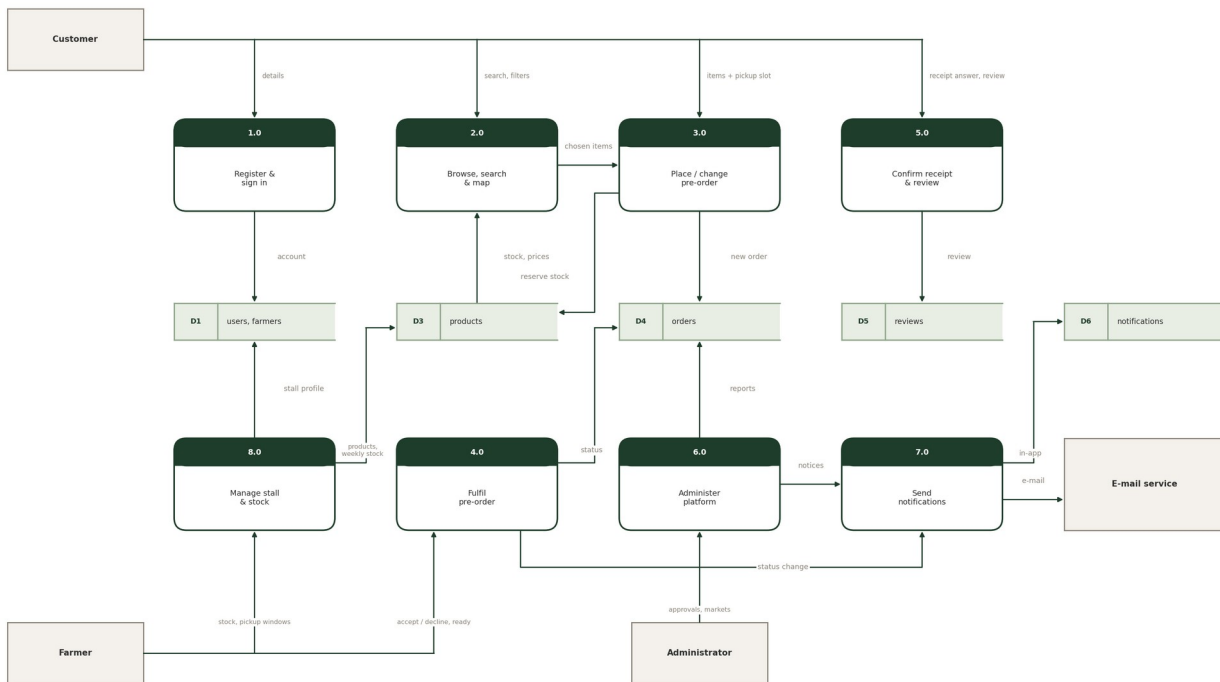


Figure 4 Data flow diagram, level 1

PROCESS	READS	WRITES
1.0 Register & sign in	Registration details	D1 users
2.0 Browse, search & map	D3 products, market and farmer data	—
3.0 Place / change pre-order	Chosen items and pickup slot	D3 stock reservation, D4 orders
4.0 Fulfil pre-order	Farmer decisions	D4 order status
5.0 Confirm receipt & review	Receipt answer, rating, comment	D5 reviews
6.0 Administer platform	D4 for reports	Approvals, markets, categories
7.0 Send notifications	Events from 4.0 and 6.0	D6 notifications, e-mail
8.0 Manage stall & stock	Stock, pickup windows	D1 stall profile, D3 products

5.3 Pre-order & pickup slot flow

FROM BROWSING TO A RESERVED SLOT

MarketLink: Customer Pre-order Flow

Browse, pre-order, pick up at the market and pay at the stall. This matches the live system: no online payment.

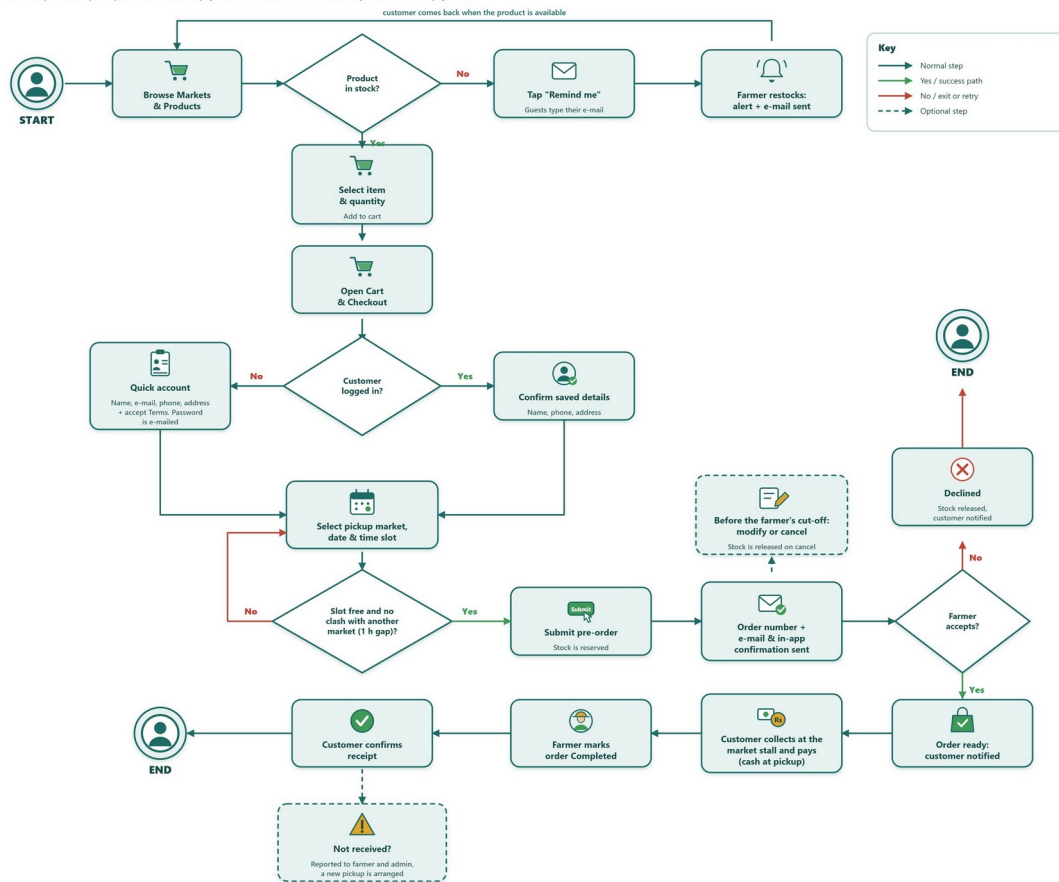


Figure 5 Pre-order and pickup slot flow

STEP	WHAT HAPPENS IN THE SYSTEM
Out of stock	The product shows "Remind me". Guests type an e-mail address; when the farmer restocks, a notification and an e-mail go out.
Checkout	Logged-in customers confirm their saved details. Guests get a quick account (name, e-mail, phone, address, Terms accepted) and the password is e-mailed.
Pickup slot	A slot is offered only if it is inside the farmer's pickup window, has free capacity and is before the cut-off. Two markets on the same day must be at least an hour apart.
Submit	Stock is reserved at once, the order number is created and a confirmation goes out by e-mail and in the app. Until the cut-off the order can be changed or cancelled.
After the order	The farmer accepts or declines, marks it ready, the customer collects and pays cash, the farmer marks it completed and the customer confirms receipt.

5.4 Order status lifecycle

SIX STATES

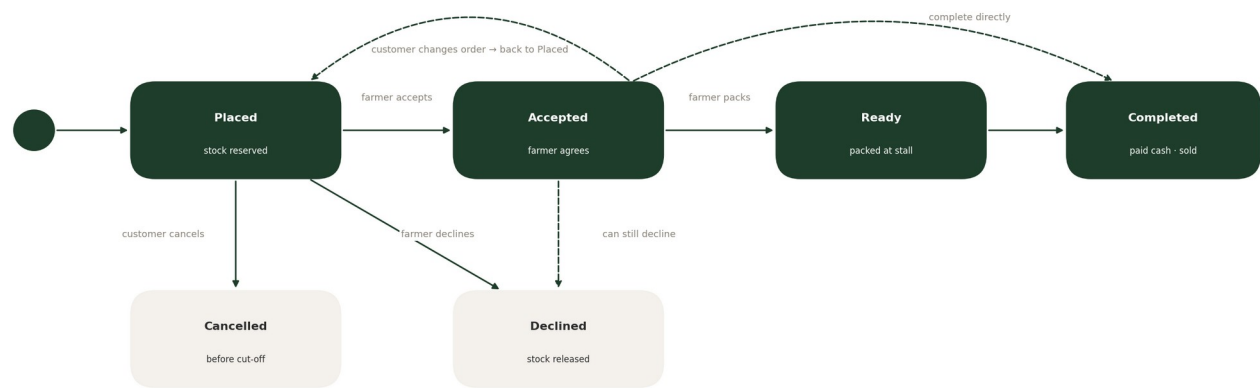


Figure 6 Order status lifecycle

STATUS	MEANING	WHO MOVES IT	STOCK
Placed	The customer sent the pre-order; the farmer has not answered yet	Customer	Reserved
Accepted	The farmer agrees to fulfil the order	Farmer	Reserved
Ready	Items are packed and waiting at the stall	Farmer	Reserved
Completed	The customer collected the order and paid in cash	Farmer	Sold
Declined	The farmer cannot fulfil the order (also after accepting)	Farmer	Released
Cancelled	The customer cancelled before the cut-off	Customer	Released

DASHED PATHS

A farmer can decline an order that was already accepted, and can complete an accepted order directly when nothing needs packing. Any change the customer makes to a placed or accepted order puts it back to Placed so the farmer has to accept it again.

5.5 Farmer registration and approval

A NEW STALL STAYS LOCKED UNTIL APPROVED

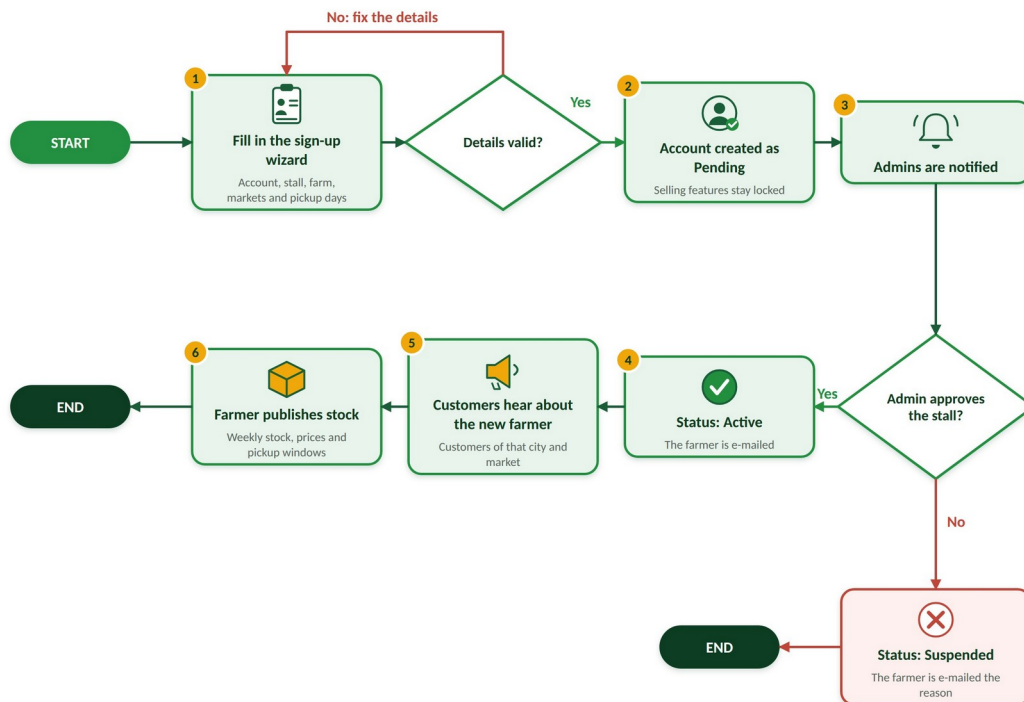


Figure 7 Farmer registration and approval

STEP	WHAT HAPPENS
1	The farmer fills in the sign-up wizard: account, stall, farm details, markets and pickup days. Invalid details send the farmer back.
2	The account is created with status Pending; selling features stay locked.
3	All administrators are notified and review the stall.
4	Approve: status becomes Active and the farmer is e-mailed. Not approved: status becomes Suspended with the reason.
5	Customers of that city and market hear about the new farmer.
6	The farmer publishes weekly stock, prices and pickup windows.

5.6 How a farmer handles a pre-order

DECIDE, PACK, COMPLETE

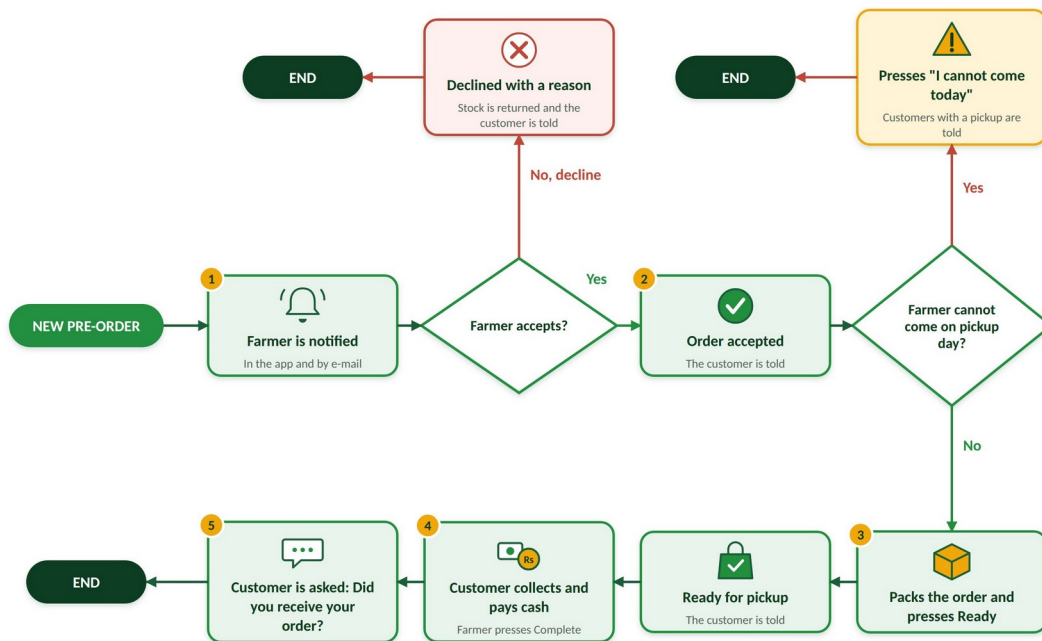


Figure 8 How a farmer handles a pre-order

STEP	WHAT HAPPENS
1	The farmer is notified in the app and by e-mail when a pre-order arrives.
2	Accept: the customer is told. Decline: the farmer gives a reason, the stock is returned and the customer is told.
3	If the farmer cannot come on the pickup day, "I cannot come today" tells every customer with a pickup that day.
4	Otherwise the farmer packs the order and presses Ready; the customer is told.
5	The customer collects and pays cash; the farmer presses Complete.
6	The customer is asked "Did you receive your order?" (Figure 9).

5.7 Receipt confirmation and review

ONLY CONFIRMED PICKUPS CAN BE REVIEWED

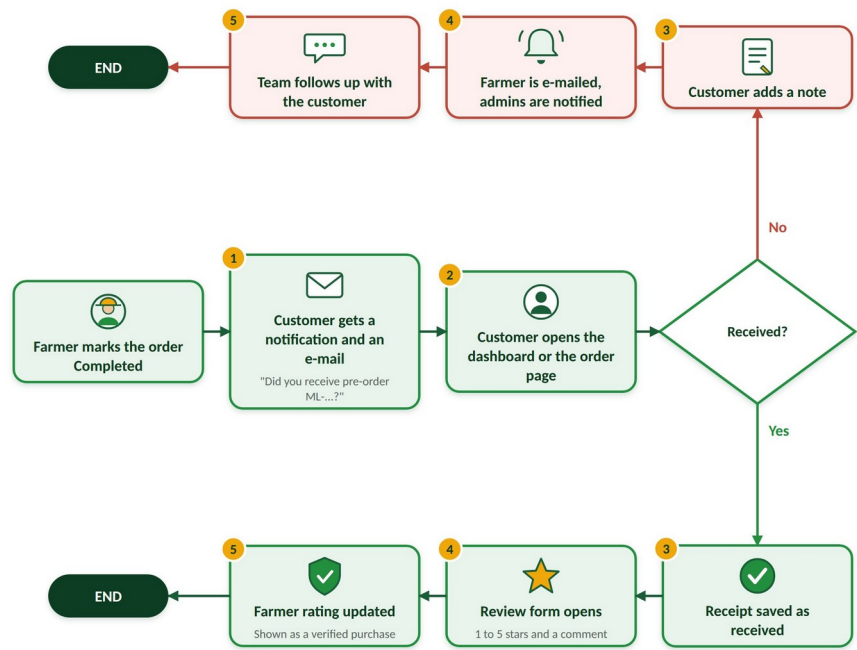


Figure 9 Receipt confirmation and review

STEP	WHAT HAPPENS
1	The customer receives a notification and an e-mail asking "Did you receive pre-order ML-...?".
2	The customer opens the dashboard or the order page and answers.
3	Yes: the receipt is saved as received. No: the customer adds a note.
4	Yes: the review form opens (1 to 5 stars and a comment). No: the farmer is e-mailed and the admins are notified.
5	Yes: the farmer rating is updated and the review is shown as a verified purchase. No: the team follows up.

5.8 How the assistant answers a question

KEYWORDS, MEMORY, LIVE DATA

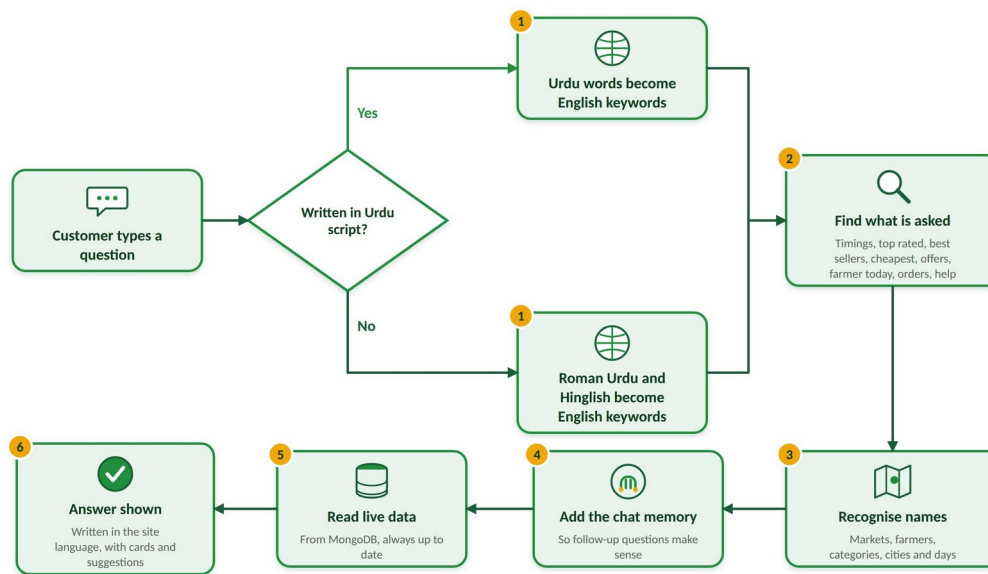


Figure 10 How the assistant answers a question

STEP	WHAT HAPPENS
1	Detects Urdu script or Roman Urdu / English and turns the words into English keywords.
2	Works out what is asked: timings, top rated, best sellers, cheapest, offers, is a farmer at the market today, order status or help.
3	Recognises the markets, farmers, categories, cities and days named in the question.
4	Adds what the chat is about from memory, so follow-up questions make sense.
5	Reads live data from MongoDB, so answers are always current.
6	Writes the answer in the site language, with cards and suggestions.

SECTION

06

Testing & test data

Thirteen build rounds took MarketLink from a basic catalogue to a complete market platform, with every round checked by automated tests before the next began.

410

API TESTS

712

BROWSER CHECKS

18

SCREEN WIDTHS

100%

PASS RATE

6.1 Test data used in the project

INPUTS, EXPECTED RESULTS AND OUTCOMES FOR THE KEY RULES

All tests run on the demo database created by the seed script (Appendix E), which is reset before every suite. The cases below use that data.

ID	SCENARIO	TEST DATA	EXPECTED	SUITE	RESULT
TC-01	Two customers want the last kilo	Product stock 1 kg; two orders of 1 kg sent one after the other	First reserved, second refused: not enough stock	smoke	Pass
TC-02	Pickup clash	Same Sunday: Clifton 10:00 and DHA Phase 6 10:30	Refused with the 60-minute message	r12clash	Pass
TC-03	Same market, same time	Two farmers, both at Clifton, both 10:00	Accepted	r12clash	Pass
TC-04	Change after cut-off	Placed order whose farmer cut-off has passed	Change refused	audit3	Pass
TC-05	Full slot	Slot at its capacity; one more order for it	Refused: slot full	audit3	Pass
TC-06	Closed date	Pickup on a date the farmer marked closed	Refused	smoke2	Pass
TC-07	Pending farmer lists a product	Farmer account with status Pending	403, product not created	audit3	Pass
TC-08	Wrong role	Customer calls an /admin route; no login on /orders	403; 401	smoke	Pass
TC-09	Guest with an existing e-mail	Guest checkout using a registered e-mail	Asked to log in	r5api	Pass
TC-10	Buying by weight	Chillies: step 250 g, min 250 g, max 2 kg; orders of 500 g and 100 g	500 g accepted, 100 g refused	r13api	Pass
TC-11	Too many farmers	Basket with items from 11 farmers	Refused: at most 10 per checkout	smoke	Pass

ID	SCENARIO	TEST DATA	EXPECTED	SUITE	RESULT
TC-12	Remind me when in stock	Restock request on a product with stock	Refused	r10test	Pass

6.2 Round-by-round delivery

WHAT EACH ROUND ADDED

ROUND	WHAT WAS ADDED	CHECKS
1-2	Foundations: server and React app, accounts and login for every role, database models, demo data; core customer, farmer and admin features	smoke, audit3, apitools, api4, e2e
3	Footer, profile photo upload, admin shell, placing an order for a customer, CSV export	Footer, avatar, admin UI
4	Password eye button, shared page shells, verified reviews, product gallery, inventory, sales insights, moderation, phone layouts	112 browser
5	One login page for all roles, search inside dropdowns, quick view, basket sidebar, checkout without an account	45 browser + 25 API
6	Home banner, video tour, review wall, newsletter, seasonal announcements, DataTables everywhere	59 browser + 44 API
7	Real product photos, banner pictures, amount dialog, empty basket, FAQs, SEO on every page	73 browser + 63 API
8	Basket speed, review wall, Urdu on every page, guest pre-order in Urdu, assistant memory	39 browser + 65 API
9	Banner and toasts, category photos, offers, FAQ accordion, filter sidebar	60 browser
10	Assistant in three languages, offer banner, photo credits, restock reminders, receipt question, "away today", farmer rankings, live search	88 browser
11	Farmer offers with a last day, market-day check-in, new home banner, 320 px phones	136 browser
12	Pickup problems, absence reasons and attendance report, unsaved-changes warnings, "Write in Urdu", reports page, clash message	76 browser + 5 API
13	Buying by weight, the same produce at other stalls, checkout items list, final polish and documents	24 browser + 20 API

6.3 Testing results

SUITES AND TOTALS

API SUITE	WHAT IT COVERS	CHECKS	PASSED
smoke	Accounts, markets, farmers, products, basket rules, pre-orders, reviews, favourites, family, role checks	69	69
smoke2	Filters, search, closed dates, reviews, maps data	23	23
audit3	SRS checklist: registration fields, near me, cut-off, slot capacity, suspend / approve	32	32
apitools	Admin tools: cities, markets, categories, accounts, place order, tables	33	33
api4	Inventory, low-stock alerts, sales insights, moderation, reports	31	31
r5api	Single login, guest checkout, verified reviews, SEO heads	25	25
r6api	Announcements by season, newsletter, FAQs, product SEO fields	44	44
r7api	FAQs admin, structured data, llms.txt, robots, sitemap	63	63
r8api	Urdu content in the API, server messages in Urdu, assistant memory	65	65
r12clash	No pickups at two markets at the same time	5	5
r13api	Buying by weight, the same produce at other stalls	20	20

API SUITE	WHAT IT COVERS	CHECKS	PASSED
Total		410	410

BROWSER & STATIC	WHAT IT COVERS	RESULT
r4test – r9test	Shells, reviews, gallery, inventory, banner, basket, FAQs, Urdu on every page	388 passed
r10test	Assistant in 3 languages, offer banner, credits, reminders, receipt, away today, rankings	88 passed
r11test	Farmer offers, market day, new home banner, 320 px phones, Urdu	136 passed
r12test, r12unsaved, r12clashui	Pickup problems, attendance, unsaved-changes warnings, Write in Urdu, clash message	76 passed
r13ui	Price comparison, buying rules, weight in basket and checkout	24 passed
resp8 (EN and UR)	Every page at 18 widths, 320 to 1920 px	567 page views each, no problems
dtsweep, align, emojiican, sw	Every DataTable, alignment, no sideways scrolling at 360 / 390 / 768 / 1024 px	No problems
Static checks	ESLint on server and client; production build; missing Urdu translations	0 errors, 0 missing

6.4 Defects found and fixed

TWENTY REAL PROBLEMS THE TESTS CAUGHT

#	DEFECT	FOUND BY	FIX
1	Saved phone number blanked the customer profile	Test r10	Phone box uses the typed number until its rules load
2	Receipt pop-up covered the dashboard for every test user	Regression	Expected for the demo customer; r10 checks it
3	Basket side panel shook when it opened	Manual	Page keeps space for the scroll bar
4	Notification menu went off the left edge on phones	Responsive	Full-width menu under the bell on phones
5	Assistant read "mera naam kya hai" as a new name	Assistant test	Roman Urdu phrase for "what's my name" added
6	"Top rated fruits in Karachi" matched a farmer named Karachi...	Assistant test	A city name alone no longer picks a farmer
7	Map tiles blocked (HTTP 403)	Manual	Correct referrer policy and a second tile server
8	Home banner did not change by itself	Manual	Autoplay timer fixed
9	Inventory page failed on an older database	API test	Query rewritten for the stored data
10	992–1199 px: account menu pushed 69 px off screen	Responsive	Links show from 1200 px; sidebar below that
11	Markets page on phones: buttons ran off the edge	Responsive	Buttons wrap onto a second line
12	Urdu top bar at 768 px was 6 px too wide	Responsive	Smaller padding in Urdu
13	Offer banner label style changed the "% off" tag	Test r9	Banner label has its own class
14	Toast title after guest checkout stayed English on Urdu site	Test r8	Urdu text added
15	Approved farmers could not say they could not come	Customer report	"Your market day" card on the dashboard

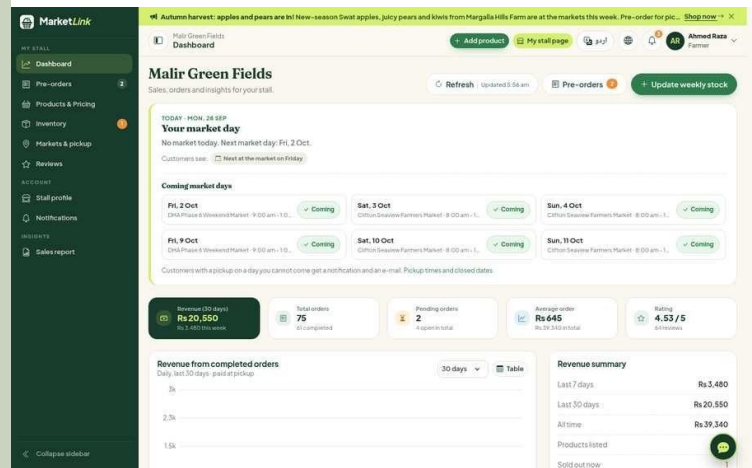
#	DEFECT	FOUND BY	FIX
16	Offer button made stock table actions too wide on phones	Test r11	Only the tag icon shows on phones
17	320 px: signed-in top bar about 30 px too wide	Responsive	Smaller buttons, gaps and logo below 375 px
18	320 px: Add / Remind me stuck out of the card	Responsive	Price may wrap; label hides
19	Assistant missed Roman Urdu "how" questions	Test r11	Plain "how" and Urdu phrasing lead to help
20	"I cannot come today" wrapped on a 390 px phone	Alignment	Slightly smaller font on phones

SECTION

07

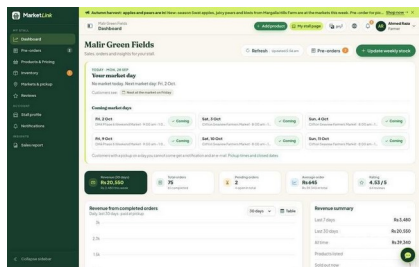
The product in pictures

Every screen below comes from the final build with the demo data loaded.

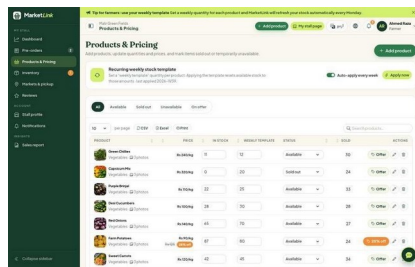


7.1 Farmer portal

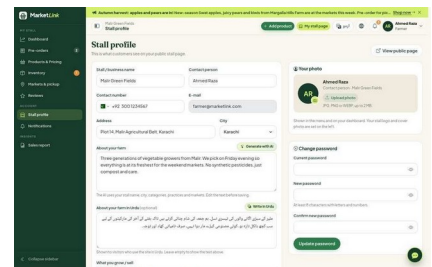
STOCK, PRICES, STALL



A Farmer dashboard



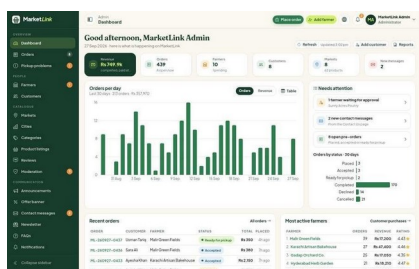
B Products & pricing



C Stall profile

7.2 Admin control centre

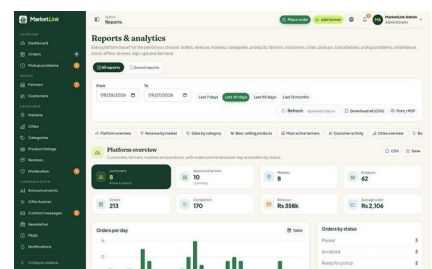
OVERSIGHT AND REPORTS



A Admin overview

Farmer	Name	Address	Status
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active
Malir Green Fields	Malir Green Fields	Malir Green Fields	Active

B Farmers table + CSV



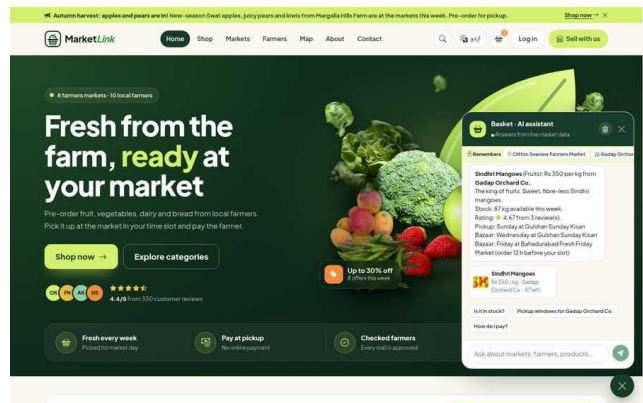
C Reports & analytics

7.3 Urdu & AI assistant

RIGHT-TO-LEFT AND ROMAN URDU



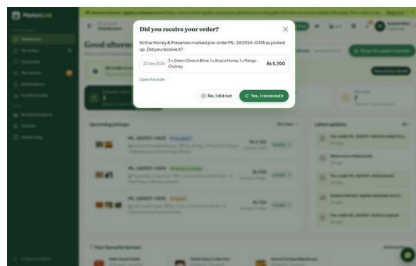
A Full Urdu page



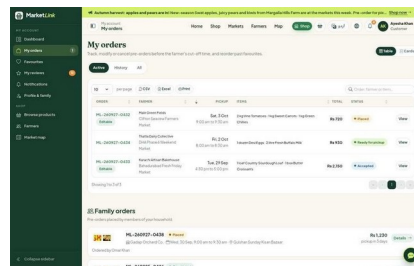
B Assistant in Roman Urdu

7.4 The customer journey

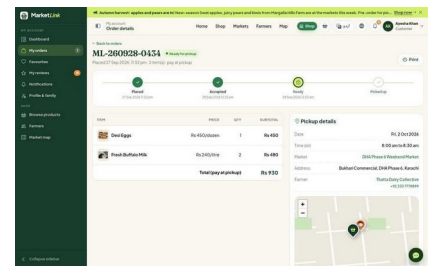
FROM THE DASHBOARD TO THE RECEIPT QUESTION



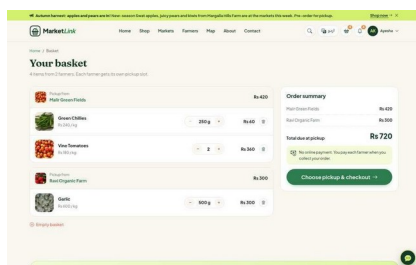
A Customer dashboard



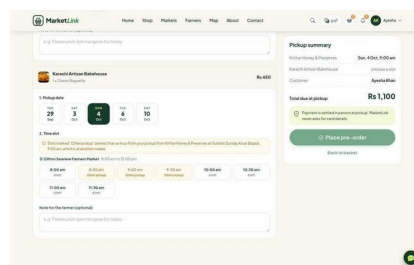
B My orders



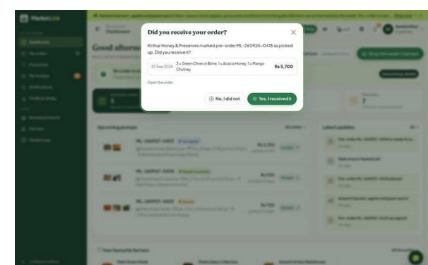
C Order with pickup details



D Buying by weight



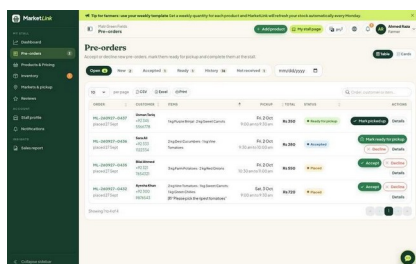
E Pickup clash message



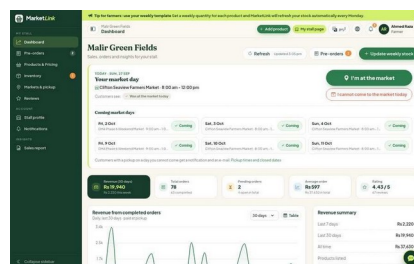
F Receipt question

7.5 Farmer and admin workflows

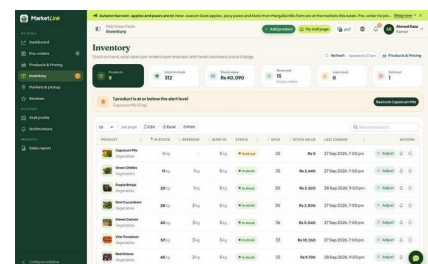
THE BAD DAYS, HANDLED



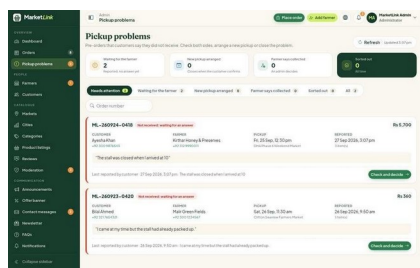
A Farmer pre-orders



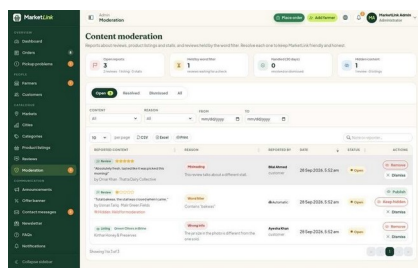
B Market-day check-in



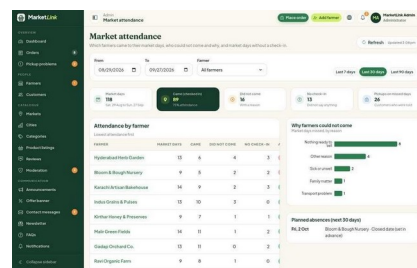
C Inventory & history



D Pickup problems



E Moderation queue



F Market attendance

7.6 Unique features that solve real problems

TWELVE PROBLEMS, TWELVE ANSWERS

#	REAL-WORLD PROBLEM	WHAT MARKETLINK DOES ABOUT IT
1	"Is the farmer even coming today?"	The farmer checks in on the market day; a live "at the market today" badge shows, and customers with an order are told when the farmer cannot come
2	Wasted trips when produce sells out early	Pre-orders reserve real stock the moment the order is placed
3	Two pickups at the same time in different places	Checkout refuses them and asks for at least an hour between
4	"I went, and my order was not there"	The customer reports it; the farmer answers; the admin verifies and decides
5	Buying chillies or ginger by the kilo is too much	The customer picks 250 g, 500 g or 1 kg within the farmer's limits
6	Same produce, different prices at several stalls	The product page shows "Rs 30 less per kg" or "Rs 20 more per kg"
7	Farmers forget to update stock every week	A weekly template refills the stall; low-stock alerts warn the farmer
8	Customers miss the product they wanted	"Remind me" e-mails the customer (or a guest) when it is back
9	Reviews that nobody trusts	Only customers who confirmed a pickup can review; verified badge; farmers reply
10	Many customers do not read English well	The whole site in Urdu, "Write in Urdu" for farmers, an assistant that understands Roman Urdu
11	Market organisers have no numbers	Seventeen reports, farmer rankings and a market attendance report
12	New customers give up at the sign-up form	Guest checkout: the account is created and the password e-mailed

7.7 Reports & analytics

SEVENTEEN REPORTS ON ONE PAGE

Each report has a date range, a chart, a table, CSV / Excel / print export and saved copies. Farmers get their own printable sales-insights report.

01	Orders summary	10	Busiest pickup days and times
02	Revenue by market	11	Cancelled and declined orders
03	Most active farmers	12	Pickup problems (not received)
04	Sales by category	13	Farmer market attendance
05	Customer activity	14	Offers and discounts
06	Inventory and low stock	15	Farmer sign-ups and approvals
07	Cities overview	16	Customer demand (remind me)

08	Reviews and moderation	17	Newsletter, messages and sign-ups
09	Best-selling products		

SECTION

08

Challenges & lessons

The hardest parts were not the pages themselves but the real-world rules around them: time, weight, language and trust between strangers.

CHALLENGE	HOW WE SOLVED IT	WHAT WE LEARNED
Two customers reserving the last kilo	Stock is decreased with one atomic database operation that fails when the stock is not enough; the order is created only after that	Business rules belong on the server and in the database
Clashing pickups	The server compares every new pickup with the customer's other slots and refuses two markets at the same time	Think about what happens between farmers, not only inside one order
Selling by weight	Farmers set the smallest step, a minimum and a maximum; quantities are rounded to three places; every screen shows "250 g" or "1.5 kg"	Amounts, money and rounding need one shared set of helpers
Urdu right-to-left	Direction-aware styles, an Urdu font, a translation file checked by a script, and "Write in Urdu"	Plan for both directions from the start
Orders not received	Cash on pickup needs a fair way to settle disputes: the customer reports, the farmer answers, the admin decides	Real markets need features for the bad days too
Farmers who cannot come	A market-day check-in, "I cannot come" with a reason, customer notices and an attendance report	Tell people early when plans change
Maps without a paid key	OpenStreetMap tiles with a fallback server, OSRM for routes, Google Maps links for directions	Free services work well when you handle their limits
Photos we may use	Open Images Dataset (CC BY 2.0) with a credits page; larger banners made with a super-resolution model	Licences and credits are part of the work
E-mail at the demo	Three modes: console (default), a free test inbox, or real SMTP	Make the default work with no setup
Lost edits	Save and Cancel highlight, and dialogs warn about unsaved changes	Protect the user's work
Speed without extra setup	Redis is used when configured; memory otherwise	Optional parts need a safe fallback
Testing a large app quickly	Suites reseed the database and run in a real browser at many widths	Automated tests find phone-width problems nobody checks by hand

Lessons learned

FOUR WE WILL KEEP

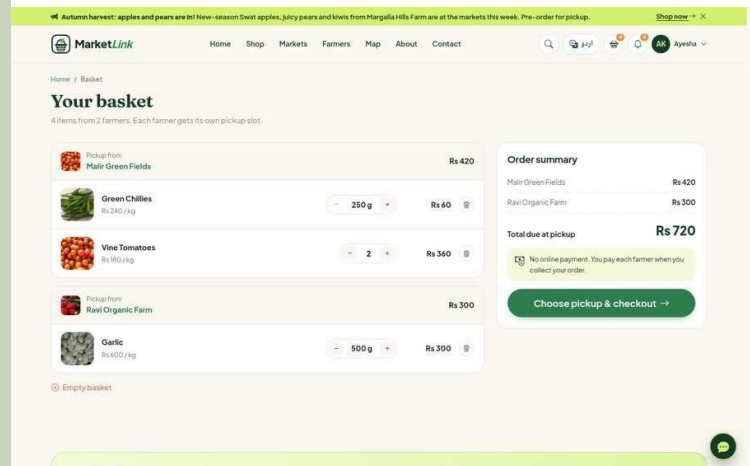
1 Design for the real market Rules such as cut-off times and pickup windows mattered more than extra screens.	2 Test every round Automated API and browser tests let us change a lot each round without breaking earlier work.
3 Language is a feature Right-to-left Urdu had to be planned into the design system, not added at the end.	4 Plan for missing services Optional parts like Redis and SMTP need a safe fallback so the app still runs anywhere.

SECTION

09

Conclusion & next steps

MarketLink delivers a complete end-to-end web solution for local farmers markets in Pakistan.



By combining online pre-orders with cash on pickup, live stock, full Urdu support and an AI assistant, MarketLink helps farmers plan their harvest and sell more of it, and gives city families a reliable way to get fresh produce. The application was built as a MERN multi-tier system, tested with more than a thousand automated checks, and is delivered as a ready-to-run package with demo data and documentation.

3

PORTALS

20

COLLECTIONS

17

REPORTS

13

ROUNDS

Next steps

WHERE MARKETLINK GOES NEXT

**Mobile app**

A mobile app for farmers to update stock straight from the market, built on the same API.

**SMS & WhatsApp reminders**

Pickup reminders for customers without regular e-mail access.

**More cities and languages**

Add markets in more cities and invite their farmers; add Sindhi and Pashto.

**Optional online payment**

A deposit option for large pre-orders while keeping cash on pickup.

**Delivery through a partner**

For customers who cannot come to the market.

**Farmer verification badges**

Identity, organic or food-safety certificates checked by the admin.

SECTION

A

Installation

Requirements: Node.js 20 or newer, and MongoDB (Community Server on the computer, or a free MongoDB Atlas cluster). Command-line details and all settings are in ReadMe.doc.

A.1 Run it on a computer

SEVEN STEPS

1. Open a terminal in the MarketLink folder.
2. Install all packages with the install command listed in the ReadMe.
3. Copy the example settings file in the server folder and fill in the MongoDB connection address and a long random secret for sign-in tokens.
4. (Optional) Create the collections and indexes by running the schema script with mongosh.
5. Load the demo data with the seed command. This deletes any existing MarketLink data.
6. Start the app in development mode and open it in the browser at localhost port 5173.
7. Or build it and start one server, then open localhost port 5000.

E-mail. Without SMTP settings the e-mails are printed in the server window. To send real e-mails, add the SMTP details to the settings file and run the mail test command.

Urdu. Press the Urdu button in the top bar, or add ?lang=ur to any page address.

A.2 Putting the site online

HOSTED ON RENDER

The repository contains a Render blueprint. It creates one web service that serves both the API and the built React app, with a health check on /api/health. A free Redis instance caches public pages and reports. The MongoDB Atlas connection address is pasted into the settings; the token secret is generated automatically and the time zone is set to Asia/Karachi.

ITEM	VALUE
Hosted URL	[paste the live URL]
Repository / zip	FBA_Omniverse.zip

SECTION

B

Requirement traceability

Every requirement of SRS sections 1.6 and 1.7, where it lives in MarketLink, and how it was checked.

CUSTOMER

ID	SRS REQUIREMENT	WHERE IN MARKETLINK	VERIFIED BY
C-01	Registration with name, contact number, e-mail, address	/register, /login; phone box with country flag	audit3 (fields)
C-02	Favourite farmers and products	Heart buttons; Account → Favourites	smoke
C-03	Family account sharing (optional)	Account → Profile → Family	smoke
C-04	Browse markets by location and day; farmers at each	/markets (open today first), market page	audit3 (near me)
C-05	Farmer profile: stall, location, days, weekly stock	/farmers/<name>	smoke
C-06	Map with markers and directions	/map, market and farmer pages (OSM, OSRM, Google Maps link)	smoke2 (map data)
C-07	Categories and filters: price, category, market, day	/products filters, /categories	smoke2
C-08	Product details: price, unit, quantity, farmer	/products/<name>	smoke
C-09	Basket and pre-order against stock	Basket, /checkout	smoke, TC-01
C-10	Pickup date and slot	/checkout slot picker	audit3, r12clash, TC-02-06
C-11	Order status; cancel or modify before cut-off	Account → Orders → order page	audit3, TC-04
C-12	Order history and reorder	Account → Orders, "Order again"	e2e journeys
C-13	Favourites and restock alerts	"Remind me" on sold-out products	r10test, TC-12
C-14	Saved markets, route-friendly pickup details	Favourites → Saved markets; order page directions	Manual walkthrough
C-15	AI assistant (optional)	Chat button on every page	r8api, r10test
C-16	Reviews after completion; visible before ordering	Receipt question, review form, product and farmer pages	r5api (verified)

FARMER

ID	SRS REQUIREMENT	WHERE IN MARKETLINK	VERIFIED BY
F-01	Registration: stall, contact, number, e-mail, address	/register/farmer wizard	audit3
F-02	Profile: markets, days, pickup windows, map pin	Farmer → Profile, Pickup	smoke2 (closed dates)
F-03	Add / edit / view / delete products	Farmer → Products & Pricing	smoke

ID	SRS REQUIREMENT	WHERE IN MARKETLINK	VERIFIED BY
F-04	Recurring weekly stock template	Farmer → Inventory; hourly job	Manual walkthrough
F-05	Mark sold out / unavailable	Farmer → Inventory	api4
F-06	Accept, decline, mark ready	Farmer → Orders	e2e journeys
F-07	Cut-off times and pickup slots	Farmer → Pickup	audit3, TC-04/05
F-08	Sales history, best sellers, orders, revenue	Farmer → Dashboard, Sales	api4 (insights)
F-09	Reply to reviews	Farmer → Reviews; own stall page	r4test

ADMINISTRATOR

ID	SRS REQUIREMENT	WHERE IN MARKETLINK	VERIFIED BY
A-01	Admin login and dashboard with totals	/admin	apitools
A-02	Approve / suspend farmers	Admin → Farmers	audit3, TC-07
A-03	Activate / deactivate customers	Admin → Customers	apitools
A-04	Manage markets with map position	Admin → Markets	apitools
A-05	Remove inappropriate listings / reviews	Admin → Moderation, Products, Reviews	api4
A-06	Reports: orders, revenue by market, active farmers	Admin → Reports, rankings	api4, r12test
A-07	Categories and announcements	Admin → Categories, Announcements	apitools, r6api

OTHER FUNCTIONAL

ID	SRS REQUIREMENT	WHERE IN MARKETLINK	VERIFIED BY
O-01	Role-based access	Role check on every protected request	smoke, TC-08
O-02	Search, sort, filter with map results	Shop, farmers, markets, map pages	smoke2
O-03	Responsive design	Every page, phone bottom bar	resp8 (18 widths)
O-04	E-mail / in-app notifications	Bell menu, Notifications page, e-mails	e2e journeys
O-05	About Us	/about	resp8
O-06	Contact Us with map	/contact	r10test (topics)

NON-FUNCTIONAL (SRS 1.7)

REQUIREMENT	HOW IT IS MET	VERIFIED BY
Compatibility	Standard HTML, CSS and JavaScript for current Chrome, Edge, Firefox and Safari	Chromium automated; others Manual walkthrough
Responsive design	18 widths from 320 to 1920 px; bottom bar and bottom-sheet dialogs on phones	resp8
Security	Hashed passwords, httpOnly cookies, server role checks, rate limits, upload checks	smoke, TC-08
Performance	Server-side paging, compression, lazy-loaded pages, sized images, optional Redis	Manual walkthrough
Usability	Toasts for every action, button spinners, empty states, unsaved-changes warnings	r12unsaved
Accessibility	Keyboard access, visible focus, alt text, reduced motion, contrast	Manual walkthrough
Languages	English and Urdu on every public, customer and farmer page	r8api, translation scan
Search engines	Titles and descriptions per page, sitemap, structured data	r7api

SECTION

C

User credentials

Demo accounts for every role and every account state, created by the seed script.

ROLE	STATE	NAME	LOGIN E-MAIL	PASSWORD
Administrator	Active	MarketLink Admin	admin@marketlink.com	Admin@123
Customer	Active	Ayesha Khan	customer@marketlink.com	Customer@123
Customer	Deactivated	Kashif Javed	inactive.customer@marketlink.com	Customer@123
Farmer	Approved	Malir Green Fields	farmer@marketlink.com	Farmer@123
Farmer	Waiting for approval	Sunny Acres Poultry	pending.farmer@marketlink.com	Farmer@123
Farmer	Suspended	Old Town Goat Dairy	suspended.farmer@marketlink.com	Farmer@123

NOTE FOR JUDGES

A deactivated customer and a suspended farmer cannot sign in; the waiting farmer can sign in but cannot sell until approved. These accounts are included so the locked states can be seen.

SUBMISSION CHECKLIST (SRS 1.9)

ITEM	WHERE
Zip of the consolidated project	Submission folder
ReadMe.doc with assumptions	Submission folder; Appendix D
Database scripts (mongosh + JSON) and the .sql note	database folder; Section 4.2
Demo video (.mp4) covering every functional requirement	Submission folder
Installation instructions	Appendix A
User credentials	Appendix C
Test data used in the project	Section 6.1
Problem definition, design specifications, diagrams	Sections 1, 3, 4, 5

SECTION

D

Assumptions, limits & AI tools

What we assumed where the SRS was silent, what the system does not do yet, and which AI tools were used.

D.1 Assumptions

ALSO LISTED IN README.DOC

- Payment is made in person at pickup; there is no payment gateway (SRS 1.5). Pickup only; no delivery.
- Farmer identity and organic or food-safety claims are not verified; tags such as "Pesticide-free" are the farmer's own description.
- All markets use one time zone (Asia/Karachi by default) for pickup slots, cut-off times and "today".
- The market-day check-in is the farmer's own word; the location is not checked.
- An offer with a last day ends after that day; the hourly job puts the usual price back.
- A customer needs at least one hour between pickups at different markets on the same day.
- Products priced per kg or per litre can be sold by weight when the farmer allows it; the smallest amount is also the step.
- "Not received" is the customer's word; the farmer can answer and the admin decides.
- Public pages and reports may be cached for up to a minute; every change empties the cache.
- The e-mail address is the login name.
- The project uses MongoDB, so the database is delivered as a mongosh script and JSON sample data instead of .sql files.
- OpenStreetMap is used instead of the paid Google Maps API; Google Maps links are offered for directions.
- Stock photos come from the Open Images Dataset under CC BY 2.0; photographers are listed on the Photo credits page.

D.2 Known limitations

HONEST EDGES

- Tests ran in Chromium only; Firefox and Safari were not tested automatically.
- Real e-mail delivery depends on the SMTP settings; the tests use a local SMTP server.
- Maps need an internet connection for the tiles; directions use the free OSRM service.
- The assistant is rule-based; very unusual questions get a polite "I couldn't find an answer" with suggestions.

D.3 AI tools used

REQUIRED BY THE SRS

TOOL	WHAT IT WAS USED FOR
Claude Code (Anthropic)	Coding assistant during development and testing; first draft of documents
Canva	Logo design
EDSR super-resolution model (OpenCV)	Enlarging the 1024 px banner photos to 1920 px for large screens

SECTION

E

Demo data

Created by the seed script and used for every test in Section 6.

DATA	RECORDS	DETAILS
Users	21	1 administrator; 12 farmers (10 approved, 1 waiting, 1 suspended); 8 customers (7 active, 1 deactivated)
Cities	9	Karachi, Lahore, Islamabad, Rawalpindi, Hyderabad, Faisalabad, Multan, Peshawar, Quetta
Markets	8	6 in Karachi, 1 in Lahore, 1 in Islamabad
Categories	8	Vegetables, Fruits, Dairy & Eggs, Baked Goods, Herbs & Greens, Honey & Preserves, Grains & Pulses, Flowers & Plants
Products	62	60 in stock, 2 sold out; 8 on offer; some sold by weight; prices Rs 50 to Rs 3,200
Pre-orders	435	Completed, cancelled, declined and upcoming orders (placed, accepted, ready)
Reviews	341	188 farmer reviews and 153 product reviews, most with a verified purchase
Stock movements	81	Restocks, weekly template runs, corrections and order reservations

MARKETS IN THE DEMO DATA

MARKET	CITY	DAYS	HOURS
Clifton Seaview Farmers Market	Karachi	Sunday, Saturday	07:00–13:00
DHA Phase 6 Weekend Market	Karachi	Sunday, Friday	08:00–14:00
Gulshan Sunday Kisan Bazaar	Karachi	Sunday, Wednesday	07:00–15:00
North Nazimabad Evening Green Market	Karachi	Wednesday, Saturday	16:00–21:00
Bahadurabad Fresh Friday Market	Karachi	Tuesday, Friday	16:00–21:00
Saddar Heritage Farmers Market	Karachi	Monday, Thursday, Saturday	06:00–12:00
Model Town Kisan Market	Lahore	Sunday, Thursday	07:00–13:00
F-9 Park Weekend Market	Islamabad	Sunday, Saturday	08:00–14:00